

# TOTALLY INVARIANT SUBVARIETIES OF PROJECTIVE SPACE

GPT 5.6 PRO, COMMUNICATED BY YUJIE LUO AND SHENG MENG

**ABSTRACT.** Let  $f : \mathbf{P}_{\mathbb{C}}^n \rightarrow \mathbf{P}_{\mathbb{C}}^n$  be a surjective endomorphism of algebraic degree greater than one. We prove that every integral closed subvariety  $V$  with  $f^{-1}(V) = V$  set-theoretically is a linear subspace.

**Disclosure.** We claim no credit for this work. We are sharing it solely to present a proof generated by GPT 5.6 Pro. We have simplified and verified the proof. More intuitive versions are being developed and are available [online](#).

## 1. INTRODUCTION

We work over  $\mathbb{C}$ . Let  $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$  be a surjective endomorphism of algebraic degree  $q > 1$ , so that  $f^*\mathcal{O}_{\mathbf{P}^n}(1) \simeq \mathcal{O}_{\mathbf{P}^n}(q)$ . A closed subvariety  $V \subseteq \mathbf{P}^n$  is *totally invariant* under  $f$  if  $f^{-1}(V) = V$  as sets. Such subvarieties arise as exceptional loci in backward dynamics. Coordinate linear subspaces are examples for the power map

$$[x_0 : \cdots : x_n] \mapsto [x_0^q : \cdots : x_n^q].$$

The following classical conjecture predicts that every integral totally invariant subvariety is linear; see [22, arXiv version, Conjecture 1.4].

**Linearity Conjecture.** *Let  $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$  be a surjective endomorphism of algebraic degree greater than one, and  $V \subseteq \mathbf{P}^n$  an integral closed subvariety. If  $f^{-1}(V) = V$  set-theoretically, then  $V$  is a linear subspace.*

For curves in  $\mathbf{P}^2$ , Fornæss–Sibony established degree bounds that left only a smooth conic as a possible nonlinear example [15, Proposition 4.2 and Section 4, p. 212]. Cerveau–Lins Neto ruled out this possibility and proved the plane-curve case [8, Theorem 2 and Sections 3–4]. Gurjar later gave another proof as a consequence of a ramification theorem [17]; see also [6, Remark 5]. Cerveau–Lins Neto and Beauville also settled the conjecture for smooth hypersurfaces in arbitrary dimension [8, 1]. The remaining cases therefore concern singular hypersurfaces and subvarieties of higher codimension.

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The exceptional set exists [13, Theorem 6.1]. Moreover, for a fixed endomorphism, there are only finitely many irreducible proper totally invariant Zariski-closed subsets of  $\mathbf{P}^n$  [13, Corollary 6.5]. In 2003, Dinh–Sibony gave the first correct proof of existence in the more general setting of polynomial-like maps [12, Section 3.4]. The projective-space result is recalled in their 2008 paper [13, Section 6]. In 2009, Dinh gave another proof using analytic multiplicative cocycles [11, Theorem 3.1].

In 2004, Briend–Cantat–Shishikura presented an incomplete proof of the linearity conjecture in full generality [2]. Their argument relied on Briend–Duval’s local-degree lemma [3, Section 2]. Dinh pointed out the error in this lemma in November 2006, as recorded in Briend–Duval’s 2009 erratum [4]; the error concerns the counting of multiplicities [13, p. 328]. The erratum proposes a replacement argument after passing to an iterate [4], but does not prove the linearity conjecture. The gap in Briend–Cantat–Shishikura’s proof is explicitly noted by Cantat–Xie [6, Remark 6].

For singular hypersurfaces, D.-Q. Zhang reduced the nonlinear case in  $\mathbf{P}^3$  to four possible singular cubic surfaces [25, Theorem 1.1]. Horing then proved that every totally invariant prime divisor in  $\mathbf{P}^3$  is a hyperplane [20, Corollary 1.2 and its proof], using a logarithmic-differential estimate together with earlier results, including Nakayama–Zhang’s exclusion of degree two [22, arXiv version, Section 5]. Mabed extended this method to irreducible totally invariant hypersurfaces with at most isolated singularities in  $\mathbf{P}^n$ , for every  $n \geq 2$  [21, Corollary 1.4]. More recently, Broustet–Horing proved linearity for all totally invariant prime divisors in  $\mathbf{P}^4$  and for normal totally invariant hypersurfaces in every dimension [5, Corollaries 1.5 and 1.3].

In higher codimension, Chen–Meng–Zhong proved that a smooth rationally connected projective threefold of Picard number two admitting an int-amplified endomorphism is toric [10, Theorem 1.3]. Their result implies that every blowup-equivariant smooth curve in  $\mathbf{P}^3$  for a non-isomorphic surjective endomorphism is a line [10, Corollary 1.6]. In arbitrary codimension, Chang–Luo established sharp bounds for the total degrees of subvarieties that are totally invariant under iterates of a fixed endomorphism [9, Theorem 1.3].

We prove the following theorem, which establishes the linearity conjecture in full.

**Theorem 1.1.** *Let  $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$  be a surjective endomorphism of algebraic degree greater than one, and let  $V \subseteq \mathbf{P}^n$  be an integral closed subvariety. If  $f^{-1}(V) = V$  set-theoretically, then  $V$  is a linear subspace.*

Theorem 1.1 imposes no smoothness or normality assumption on  $V$ . It also holds over any algebraically closed field of characteristic zero: one can descend  $f$  and  $V$  to a field finitely generated over  $\mathbb{Q}$  and embed that field into  $\mathbb{C}$  [24, Tag 01ZM]. Both total invariance and linearity can be checked after extending the ground field. The distinction between set-theoretic and

scheme-theoretic inverse images is central to the proof: a general fiber of the restriction to  $V$  is reduced, while the corresponding fiber in  $\mathbf{P}^n$  need not be.

Here is the argument. Write  $r = \dim V$  and  $\delta = \deg V$ , and assume  $1 \leq r < n$ . The trace splitting of [9], together with iteration, gives

$$H^i(\mathbf{P}^n, \mathcal{I}_V(t)) = 0 \quad (i > 0, t > 0).$$

Thus, for every  $t > 0$ , sections of  $\mathcal{O}_V(t)$  extend to  $\mathbf{P}^n$ , and  $H^i(V, \mathcal{O}_V(t)) = 0$  for all  $i > 0$ .

We may replace  $f$  by an iterate and continue to write  $f$  and  $q$  for the map and its algebraic degree. Put  $g = f|_V$ . A general fiber  $S = g^{-1}(y)$  consists of  $q^r$  distinct smooth points. A relative Jacobian calculation produces a single polynomial, of degree at most  $(n - r)(q - 1)$ , whose class generates the socle of the ambient fiber algebra at each point of  $S$ . Multiplying by this polynomial turns local residues into nonzero multiples of point evaluation. Euler–Jacobi vanishing then gives a relation

$$\sum_{x \in S} \lambda_x \sigma(x) = 0, \quad \lambda_x \neq 0, \quad \sigma \in H^0(V, \mathcal{O}_V(rq - r - 1)).$$

Values are taken in a fixed trivialization along  $S$ .

If  $\delta > 1$ , a general complementary linear section of  $V$  contains at least two points. Its inverse image under  $g$  is a reduced complete intersection on  $V$ . Extend the relation on one fiber by zero on the remaining fibers. Complete-intersection duality lifts the resulting relation to a nonzero section of  $\omega_V(r + 1)$  vanishing on another entire fiber. Since the canonical sheaf  $\omega_V$  is torsion-free of rank one, there is an injection

$$\omega_V(r + 1) \hookrightarrow \mathcal{O}_V(m)$$

for some  $m > 0$  depending only on  $V \subseteq \mathbf{P}^n$ . The resulting section of  $\mathcal{O}_V(m)$  vanishes at all  $q^r$  points of a fiber  $g^{-1}(y')$ . Intersecting its zero divisor with the pullbacks of  $r - 1$  general target hyperplanes through  $y'$  gives

$$q^r \leq mq^{r-1}\delta.$$

Choosing the iterate so that  $q > m\delta$  gives a contradiction, so  $\delta = 1$ .

Section 2 establishes the cohomology vanishing. Sections 3 and 4 construct the fiber relation, Section 5 proves the lifting statement, and Section 6 completes the degree comparison.

For a closed subscheme  $X \subseteq \mathbf{P}^n$ , we write  $\mathcal{O}_X(t) = \mathcal{O}_{\mathbf{P}^n}(t)|_X$  and, for a coherent sheaf  $\mathcal{F}$  on  $X$ ,  $\mathcal{F}(t) = \mathcal{F} \otimes \mathcal{O}_X(t)$ . The smooth locus of  $V$  is denoted by  $V_{\text{reg}}$ , and the dual of a finite-dimensional vector space  $E$  by  $E^\vee$ .

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## 2. TRACE AND COHOMOLOGY VANISHING

We first record the degrees of the ambient map and its restriction.

**Lemma 2.1.** *Let  $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$  be a surjective endomorphism of algebraic degree  $q$ . Then  $f$  is finite flat of degree  $q^n$ . If  $V$  is an integral totally invariant subvariety of dimension  $r$ , then  $g = f|_V$  is finite surjective and*

$$g^* \mathcal{O}_V(1) \simeq \mathcal{O}_V(q), \quad \deg g = q^r.$$

*There is a dense open set  $U \subseteq V_{\text{reg}}$  such that  $g^{-1}(U) \subseteq V_{\text{reg}}$  and  $g^{-1}(U) \rightarrow U$  is finite étale.*

*Proof.* A positive-dimensional fiber of  $f$  would contain a curve on which  $f^* \mathcal{O}_{\mathbf{P}^n}(1) = \mathcal{O}_{\mathbf{P}^n}(q)$  has degree zero, contradicting ampleness. Thus  $f$  is proper and quasi-finite, hence finite. It is flat by miracle flatness, since source and target are smooth of the same dimension [24, Tag 00R4]. The projection formula gives  $\deg f = q^n$ .

Total invariance and surjectivity give  $f(V) = V$ . Since  $V$  is reduced,  $f|_V$  factors through a finite surjective morphism  $g : V \rightarrow V$ . The polarization restricts to  $V$ , and the projection formula gives

$$(\deg g) \deg V = q^r \deg V.$$

Hence  $\deg g = q^r$ . In characteristic zero the restriction is generically étale. Removing its branch locus,  $V \setminus V_{\text{reg}}$ , and the image of  $V \setminus V_{\text{reg}}$  gives  $U$ .  $\square$

The next proposition is the trace argument of [9, Lemma 2.1] in projective space.

**Proposition 2.2.** *Let  $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$  be a surjective endomorphism of algebraic degree  $q > 1$ , and let  $V$  be a reduced closed subscheme with  $f^{-1}(V) = V$  set-theoretically. Then  $\mathcal{I}_V(t)$  is a direct summand of  $f_* \mathcal{I}_V(qt)$  for every integer  $t$ . Consequently,*

$$(2.1) \quad H^i(\mathbf{P}^n, \mathcal{I}_V(t)) = 0 \quad (i > 0, t > 0).$$

*Proof.* The finite flat morphism  $f$  has rank  $q^n$ , and its algebra trace commutes with base change [24, Tag 0BT8]. We claim that

$$\text{Tr}_f(f_* \mathcal{I}_V) \subseteq \mathcal{I}_V.$$

Indeed, let  $b$  be a local section of  $f_* \mathcal{I}_V$  near a closed point  $y \in V$ . Every point of  $f^{-1}(y)$  belongs to  $V$ . The image of  $b$  in the finite fiber algebra therefore vanishes at every point and is nilpotent. Multiplication by this image has trace zero, so  $\text{Tr}_f(b)(y) = 0$ . Since  $V$  is reduced, this proves the claim.

Pullback also preserves  $\mathcal{I}_V$ . The composite

$$\mathcal{I}_V \longrightarrow f_* \mathcal{I}_V \xrightarrow{\text{Tr}_f} \mathcal{I}_V$$

is multiplication by  $q^n$ . Dividing the trace by  $q^n$  gives a splitting. Tensoring with  $\mathcal{O}_{\mathbf{P}^n}(t)$  and applying the projection formula proves the direct-summand assertion. Iterating the splitting shows that, for fixed  $i, t > 0$ , the group  $H^i(\mathbf{P}^n, \mathcal{I}_V(t))$  is a direct summand of  $H^i(\mathbf{P}^n, \mathcal{I}_V(q^k t))$  for every  $k \geq 1$ . Serre vanishing [19, Theorem III.5.2] makes the latter group zero for  $k \gg 0$ , proving (2.1).  $\square$

**Corollary 2.3.** *Under the hypotheses of Proposition 2.2, for every  $t > 0$  the restriction map*

$$H^0(\mathbf{P}^n, \mathcal{O}_{\mathbf{P}^n}(t)) \longrightarrow H^0(V, \mathcal{O}_V(t))$$

*is surjective, and  $H^i(V, \mathcal{O}_V(t)) = 0$  for all  $i > 0$ .*

*Proof.* Apply (2.1) and the cohomology vanishing for  $\mathcal{O}_{\mathbf{P}^n}(t)$  to

$$0 \longrightarrow \mathcal{I}_V(t) \longrightarrow \mathcal{O}_{\mathbf{P}^n}(t) \longrightarrow \mathcal{O}_V(t) \longrightarrow 0. \quad \square$$

### 3. A RELATIVE JACOBIAN LEMMA

For an Artinian local ring  $(Q, \mathfrak{m}_Q)$ , its socle is  $\text{Soc}(Q) = \text{Ann}_Q(\mathfrak{m}_Q)$ . The Jacobian theorem of Scheja–Storch states that, over a field of characteristic zero, the socle of a zero-dimensional complete intersection

$$Q = K[[z_1, \dots, z_c]]/(H_1, \dots, H_c)$$

is generated by the nonzero class of  $\det(\partial H_a / \partial z_b)$ ; see [23] and [14, Theorems 7.1 and A.5]. We need a relative version in which parameters on the reduced space may be lifted arbitrarily to the thickening.

**Lemma 3.1.** *Let  $B = \mathbb{C}[[s_1, \dots, s_r]]$ ,  $R = B[[z_1, \dots, z_c]]$ , and*

$$A = R/(H_1, \dots, H_c), \quad \Delta = \det \left( \frac{\partial H_a}{\partial z_b} \right) \in A,$$

*where  $r, c \geq 1$  and the  $H_a$  form a regular sequence. Assume that  $A$  is finite flat over  $B$  and that  $A_{\text{red}} \simeq B$  as a  $B$ -algebra. If  $N$  is the nilradical of  $A$ , then*

$$\text{Ann}_A(N) = B\Delta.$$

*Moreover, if  $\tau_1, \dots, \tau_r \in A$  reduce to a regular system of parameters of  $A/N$ , then the image of  $\Delta$  generates the socle of  $A/(\tau_1, \dots, \tau_r)$  and is nonzero.*

*Proof.* Translate each  $z_b$  by its image in  $A/N = B$ . Then the kernel of  $R \rightarrow A/N$  is  $(z_1, \dots, z_c)$ , and differentiation in the  $z_b$  with the  $s_i$  fixed is unchanged. Let  $K$  be the fraction field of  $B$  and put  $\mathfrak{p} = (z_1, \dots, z_c) \subset R$ . Since  $\sqrt{(H_1, \dots, H_c)} = \mathfrak{p}$ , the finite local  $K$ -algebra  $A_K = A \otimes_B K$  identifies with  $R_{\mathfrak{p}}/(H_1, \dots, H_c)$ . Completing  $R_{\mathfrak{p}}$  gives  $K[[z_1, \dots, z_c]]$  and leaves its Artinian quotient  $A_K$  unchanged. This presents  $A_K$  as a zero-dimensional complete intersection with residue field  $K$ . The Jacobian theorem therefore

gives  $N_K \Delta = 0$ . Since  $A$  is  $B$ -flat,  $A \rightarrow A_K$  is injective, and hence  $N\Delta = 0$  in  $A$ .

Put  $M = \text{Ann}_A(N)$ . The closed fiber  $A/\mathfrak{m}_B A$  is an Artinian complete intersection and hence Gorenstein. Lift a linear functional defining its perfect multiplication pairing to a  $B$ -linear map  $\ell : A \rightarrow B$ . The  $A$ -linear map

$$A \longrightarrow \text{Hom}_B(A, B), \quad a \longmapsto (b \longmapsto \ell(ab)),$$

is an isomorphism modulo  $\mathfrak{m}_B$  and hence an isomorphism between free  $B$ -modules of the same rank. Finite adjunction consequently gives an isomorphism of  $B$ -modules

$$M = \text{Hom}_A(B, A) \simeq \text{Hom}_A(B, \text{Hom}_B(A, B)) \simeq B.$$

The Jacobian theorem on the closed fiber gives  $\Delta \notin \mathfrak{m}_B A$ , and therefore  $\Delta \notin \mathfrak{m}_B M$ . As  $M$  is free of rank one over  $B$ , it follows that  $M = B\Delta$ .

Now let  $B' = \mathbb{C}[[T_1, \dots, T_r]]$  act on  $A$  by  $T_i \mapsto \tau_i$ . The induced map  $B' \rightarrow A/N$  is an isomorphism. Each quotient in the finite filtration by powers of  $N$  is finite over  $A/N$ , so  $A$  is finite over  $B'$ . The  $\tau_i$  form a system of parameters in the Cohen–Macaulay ring  $A$ ; thus  $A$  is free over  $B'$ . The closed fiber  $A/(\tau_1, \dots, \tau_r)$  is again a zero-dimensional complete intersection, so  $\text{Hom}_{B'}(A, B')$  is free of rank one over  $A$  by the same argument.

The quotient  $A \rightarrow A/N = B'$  splits as a map of  $B'$ -modules. Dualizing gives a split injection  $B' \hookrightarrow \text{Hom}_{B'}(A, B')$  whose image consists of the functionals vanishing on  $N$ . Under an  $A$ -linear identification  $\text{Hom}_{B'}(A, B') \simeq A$ , this image is precisely  $\text{Ann}_A(N) = M$ . Thus  $M \hookrightarrow A$  splits over  $B'$ . The actions of  $B$  and  $B'$  on  $M$  both factor through  $A/N$ , so  $M = B'\Delta$ . Reduction modulo  $(T_1, \dots, T_r)$  preserves the split injection. Hence  $\Delta$  remains nonzero in  $A/(\tau_1, \dots, \tau_r)$ . The maximal ideal of this quotient is the image of  $N$  and annihilates  $\Delta$ . Its socle is one-dimensional, proving the assertion.  $\square$

#### 4. RELATIONS ON A FIBER

Let  $p_1, \dots, p_n$  be polynomials in affine coordinates  $\xi_1, \dots, \xi_n$ , with an isolated common zero  $x$ . Write the local Grothendieck residue as

$$\rho_x(P) = \text{Res}_x \left[ \frac{P d\xi_1 \wedge \dots \wedge d\xi_n}{p_1 \cdots p_n} \right].$$

The pairing  $(a, b) \mapsto \rho_x(ab)$  is perfect on the local complete-intersection algebra [7, Theorem 1.5.1]. If the highest homogeneous parts of the  $p_i$  have no common nonzero zero, Euler–Jacobi vanishing gives

$$(4.1) \quad \sum_x \rho_x(P) = 0 \quad \text{for} \quad \deg P < \sum_{i=1}^n (\deg p_i - 1).$$

The sum includes all common zeros, and the fiber algebras need not be reduced [7, Theorem 1.5.12].

**Proposition 4.1.** *Let  $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$  be a surjective endomorphism of algebraic degree  $q > 1$ , and let  $V$  be an integral totally invariant subvariety of*

dimension  $1 \leq r < n$ . Put  $g = f|_V$  and  $t = r(q-1) - 1$ . Suppose  $y \in V_{\text{reg}}$ ,  $g$  is étale over  $y$ , and  $S = g^{-1}(y) \subseteq V_{\text{reg}}$ . After choosing a trivialization of  $\mathcal{O}_V(t)$  along  $S$ , there are nonzero scalars  $\lambda_x$ ,  $x \in S$ , such that

$$(4.2) \quad \sum_{x \in S} \lambda_x \sigma(x) = 0 \quad \text{for every } \sigma \in H^0(V, \mathcal{O}_V(t)).$$

*Proof.* Put  $c = n - r$ . Choose target coordinates  $[Y_0 : \cdots : Y_n]$  with  $y = [1 : 0 : \cdots : 0]$ , and write  $u_i = Y_i/Y_0$ . We may assume that  $u_1, \dots, u_r$  restrict to parameters on  $V$  at  $y$ , while  $du_{r+1}, \dots, du_n$  span its conormal space there. The completed ideal of  $V$  at  $y$  then has generators

$$h_a = u_{r+a} + O((u_1, \dots, u_n)^2), \quad 1 \leq a \leq c.$$

In particular,  $u_1, \dots, u_r, h_1, \dots, h_c$  are ambient local parameters.

Write  $f = [\Phi_0 : \cdots : \Phi_n]$ , where the  $\Phi_i$  are degree- $q$  forms without a common zero. Choose a source hyperplane avoiding  $f^{-1}(y)$  and not contained in any  $\{\Phi_i = 0\}$  for  $1 \leq i \leq n$ . On its affine complement, let  $p_i$  be the dehomogenization of  $\Phi_i$ , for  $0 \leq i \leq n$ . Then  $\deg p_i = q$  for  $1 \leq i \leq n$ , and  $p_1 = \cdots = p_n = 0$  defines the whole ambient fiber. Their highest homogeneous parts have no common nonzero zero: such a zero would give a point at infinity with  $\Phi_1 = \cdots = \Phi_n = 0$  and  $\Phi_0 \neq 0$ , contrary to the choice of the hyperplane.

Choose affine linear coordinates  $(s_1, \dots, s_r, z_1, \dots, z_c)$  such that the  $ds_i$  restrict to a basis of the cotangent space of  $V$  at every point of  $S$ . This is a nonempty open condition for each of finitely many points, so the choices can be made simultaneously. On this affine chart, define the polynomial

$$\Theta = \det \left( \frac{\partial p_{r+a}}{\partial z_b} \right)_{1 \leq a, b \leq c}.$$

It has degree at most  $c(q-1)$ . We claim that its class generates the socle of the local ambient fiber algebra at every  $x \in S$ .

Fix  $x \in S$  and center the source coordinates at  $x$ . In the completed local ring,  $V$  is a graph  $z_b = \varphi_b(s)$ . Set  $w_b = z_b - \varphi_b(s)$ , so that differentiation in  $w_b$  agrees with differentiation in  $z_b$  when  $s$  is fixed. The function  $p_0$  is a unit at  $x$ . Put

$$v_i = p_i/p_0 = u_i \circ f, \quad H_a = h_a(v_1, \dots, v_n), \quad A = \widehat{\mathcal{O}}_{\mathbf{P}^n, x} / (H_1, \dots, H_c).$$

To apply Lemma 3.1, let  $I$  and  $J$  be the ideals of  $V$  and  $f^{-1}(V)$  in the uncompleted local ring at  $x$ . Total invariance gives  $\sqrt{J} = I$ , hence  $I^e \subseteq J \subseteq I$  for some  $e \geq 1$ . After completion this becomes

$$(w_1, \dots, w_c)^e \subseteq (H_1, \dots, H_c) \subseteq (w_1, \dots, w_c).$$

Thus  $A_{\text{red}} = B = \mathbb{C}[[s_1, \dots, s_r]]$ . The monomials in the  $w_b$  of total degree less than  $e$  generate  $A$  over  $B$ , so  $A$  is finite over  $B$ . The  $c$  elements  $H_a$  generate an ideal of height  $c$  in a regular local ring and form a regular sequence. Consequently  $A$  is Cohen–Macaulay, and the parameters  $s_1, \dots, s_r$  make it free over  $B$ .

Since  $g$  is étale at  $x$ , the reductions of  $v_1, \dots, v_r$  are parameters of  $A_{\text{red}}$ . Apply Lemma 3.1 with normal variables  $w_b$  and parameter lifts  $\tau_i = v_i$ . It follows that the relative Jacobian

$$\Delta = \det \left( \frac{\partial H_a}{\partial w_b} \right) = \det \left( \frac{\partial H_a}{\partial z_b} \right)$$

has nonzero socle class in  $A/(v_1, \dots, v_r)$ . This quotient is the local algebra of  $f^{-1}(y)$  at  $x$ , because  $u_1, \dots, u_r, h_1, \dots, h_c$  are local parameters at  $y$ . Equivalently, it is the quotient of the completed source ring by  $(p_1, \dots, p_n)$ . In this quotient the chain rule gives

$$\frac{\partial H_a}{\partial z_b} \equiv \frac{\partial v_{r+a}}{\partial z_b} \equiv p_0^{-1} \frac{\partial p_{r+a}}{\partial z_b}.$$

Indeed,  $H_a - v_{r+a} \in (v_1, \dots, v_n)^2$ , so its derivatives lie in  $(v_1, \dots, v_n)$ . Differentiating  $v_{r+a} = p_{r+a}/p_0$  gives the second congruence, since the additional term is divisible by  $p_{r+a}$ . Taking determinants yields

$$\Delta \equiv p_0^{-c} \Theta \pmod{(p_1, \dots, p_n)}.$$

Since  $p_0$  is a unit,  $\Theta$  also has nonzero socle class. The local fiber algebra is Artinian and equals its completion, so this formal calculation applies to the algebra on which  $\rho_x$  is defined. This proves the claim.

Using the fixed affine coordinates  $(s, z)$ , set  $\lambda_x = \rho_x(\Theta)$ . The socle property gives

$$\rho_x(\Theta P) = \lambda_x P(x) \quad \text{for every local function } P.$$

Moreover,  $\lambda_x \neq 0$ : otherwise the nonzero class of  $\Theta$  would pair to zero with every element, contradicting the perfect residue pairing.

Since  $c = n - r$  and  $t = r(q - 1) - 1$ , a polynomial  $P$  of degree at most  $t$  satisfies

$$\deg(\Theta P) \leq c(q - 1) + t = n(q - 1) - 1.$$

The support of the ambient fiber is exactly  $S$ , so (4.1) gives

$$0 = \sum_{x \in S} \rho_x(\Theta P) = \sum_{x \in S} \lambda_x P(x).$$

The chosen affine chart trivializes  $\mathcal{O}_V(t)$  along  $S$ . For  $t > 0$ , every section of  $\mathcal{O}_V(t)$  is represented in this chart by a polynomial of degree at most  $t$ , by Corollary 2.3. The only remaining case is  $t = 0$ , when every global section is constant because  $V$  is integral and projective. Thus (4.2) holds in all cases.  $\square$

## 5. LIFTING RELATIONS BY DUALITY

Let  $V \subseteq \mathbf{P}^n$  be integral of dimension  $r$ . Its canonical sheaf  $\omega_V$  is the degree  $-r$  cohomology sheaf of its normalized dualizing complex  $\omega_V^\bullet$ . It is a coherent torsion-free sheaf of rank one and agrees with the canonical line bundle on  $V_{\text{reg}}$  [24, Tags 0FVV and 0AXY]. In this section we use proper duality as in [18] and [24, Tag 0FVV].

The following form of complete-intersection duality turns a relation among point values into a global section. The common zero scheme lies in the smooth locus, but  $V$  itself need not be Cohen–Macaulay.

**Proposition 5.1.** *Assume  $r \geq 1$  and  $H^i(V, \mathcal{O}_V(j)) = 0$  for all  $i, j > 0$ . Let  $q > r + 1$  be an integer, and let  $\sigma_1, \dots, \sigma_r \in H^0(V, \mathcal{O}_V(q))$  have a reduced zero-dimensional common zero scheme  $T \subseteq V_{\text{reg}}$ . Put  $t = rq - r - 1$ .*

*Let  $\lambda$  be a functional on  $H^0(T, \mathcal{O}_T(t))$  annihilating the restrictions of all sections of  $\mathcal{O}_V(t)$ . Then  $\lambda$  is induced by a section*

$$a \in H^0(V, \omega_V(r + 1)),$$

*through duality on  $T$  and adjunction for the ordered sections  $\sigma_i$ . Under the decomposition*

$$H^0(T, \mathcal{O}_T(t)) = \bigoplus_{x \in T} \mathcal{O}_T(t)|_x,$$

*the restriction of  $\lambda$  to the summand at  $x$  is zero if and only if  $a(x) = 0$ .*

*Proof.* Write  $i : T \hookrightarrow V$  and set  $E = i_* \mathcal{O}_T(t)$ . For  $0 \leq j \leq r$ , put

$$K_j = \mathcal{O}_V(t - jq)^{\oplus \binom{r}{j}}.$$

The ordered sections  $\sigma_1, \dots, \sigma_r$  give the twisted Koszul resolution

$$(5.1) \quad 0 \longrightarrow K_r \longrightarrow \dots \longrightarrow K_0 \longrightarrow E \longrightarrow 0.$$

Indeed, at a point of  $T$  they generate the maximal ideal of a regular local ring of dimension  $r$ ; away from  $T$  one of them is a unit in a local trivialization. The leftmost term is

$$M = K_r = \mathcal{O}_V(t - rq) = \mathcal{O}_V(-r - 1).$$

All other terms have positive twists, since

$$t - jq = (r - j)q - r - 1 > 0 \quad (0 \leq j < r).$$

Their higher cohomology vanishes by assumption. The successive connecting maps therefore give an exact sequence

$$(5.2) \quad H^0(V, \mathcal{O}_V(t)) \longrightarrow H^0(T, \mathcal{O}_T(t)) \xrightarrow{\partial} H^r(V, M) \longrightarrow 0.$$

We show that, under duality, the transpose of  $\partial$  is restriction of sections of  $\omega_V(r + 1)$  to  $T$ . Let

$$\varepsilon \in \text{Ext}_V^r(E, M)$$

be the extension class of (5.1); its connecting homomorphism is  $\partial$ . Proper duality gives

$$\begin{aligned} H^0(V, \omega_V(r + 1)) &\simeq \text{Hom}_V(M, \omega_V) \simeq H^r(V, M)^\vee, \\ \text{Ext}_V^r(E, \omega_V) &\simeq H^0(T, \mathcal{O}_T(t))^\vee. \end{aligned}$$

We regard a section  $a$  of  $\omega_V(r + 1)$  as a morphism  $a : M \rightarrow \omega_V$  via the first identification. The first line uses top-degree duality. For the second line, the canonical morphism  $\omega_V[r] \rightarrow \omega_V^\bullet$  is an isomorphism near  $T$ . As  $E$  is supported on  $T$ , applying  $R\mathcal{H}om_V(E, -)$  gives an isomorphism, and proper

duality applies. Both duality identifications are compatible with morphisms and connecting maps [24, Tag 0FVV(5),(6)]. Hence  $\partial^\vee$  is, up to the common sign convention, the map

$$\mathrm{Hom}_V(M, \omega_V) \longrightarrow \mathrm{Ext}_V^r(E, \omega_V), \quad a \longmapsto a_*\varepsilon,$$

obtained by pushing the extension along  $a : M \rightarrow \omega_V$ .

To compute this map, apply  $\mathcal{H}om_V(-, \omega_V)$  to (5.1). Its final differential fits into the exact sequence

$$\omega_V(r+1-q)^{\oplus r} \xrightarrow{(\pm\sigma_1, \dots, \pm\sigma_r)} \omega_V(r+1) \longrightarrow i_*(\omega_V(r+1)|_T) \longrightarrow 0.$$

There are no other cohomology sheaves: near  $T$  this is the dual Koszul complex of a regular sequence on an invertible sheaf, and away from  $T$  the complex is contractible. It follows that

$$\mathrm{Ext}_V^r(E, \omega_V) \simeq H^0(T, \omega_V(r+1)|_T).$$

The class  $a_*\varepsilon$  is represented by  $a : K_r = M \rightarrow \omega_V$  in the last term of the dual complex, so its image is precisely  $a|_T$ . Dualizing (5.2) now proves the lifting assertion.

The identifications above commute with the direct-sum decomposition over the points of  $T$ . At each point they give a nondegenerate pairing between one-dimensional spaces. Consequently the component of the functional at  $x$  vanishes exactly when  $a(x) = 0$ .  $\square$

The adjunction twist  $rq - t = r + 1$  is independent of  $q$ . The following lemma embeds  $\omega_V(r+1)$  into a fixed twist of  $\mathcal{O}_V$ .

**Lemma 5.2.** *Let  $V \subseteq \mathbf{P}^n$  be integral of dimension  $r$ . There is an integer  $m > 0$ , depending only on this embedding, and an injection*

$$\omega_V(r+1) \hookrightarrow \mathcal{O}_V(m).$$

*Proof.* For  $m \gg 0$ , the sheaf  $\mathcal{H}om_V(\omega_V(r+1), \mathcal{O}_V)(m)$  is globally generated. It has rank one, so a section nonzero at the generic point gives a generically injective morphism  $\omega_V(r+1) \rightarrow \mathcal{O}_V(m)$ . The kernel is torsion and hence zero, since  $\omega_V(r+1)$  is torsion-free.  $\square$

## 6. PROOF OF THE THEOREM

We first bound the number of points in a fiber on which a fixed-degree section can vanish.

**Lemma 6.1.** *Let  $V \subseteq \mathbf{P}^n$  be integral of dimension  $r \geq 1$  and degree  $\delta$ . Let  $g : V \rightarrow V$  be finite surjective, with  $g^*\mathcal{O}_V(1) \simeq \mathcal{O}_V(q)$  for an integer  $q \geq 1$ . Suppose  $y \in V_{\mathrm{reg}}$ ,  $g$  is étale over  $y$ , and  $S = g^{-1}(y) \subseteq V_{\mathrm{reg}}$ . If a nonzero section of  $\mathcal{O}_V(m)$ ,  $m > 0$ , vanishes at every point of  $S$ , then  $q \leq m\delta$ .*

*Proof.* Let  $Z$  be the zero divisor of the section. It is an effective Cartier divisor, since  $V$  is integral. The projection formula gives  $\deg g = q^r$ , so  $S$  consists of  $q^r$  distinct points.

Pullbacks of target hyperplanes through  $y$  form a linear system in  $|\mathcal{O}_V(q)|$  with finite base locus  $S$ . Choose  $r - 1$  general members  $E_1, \dots, E_{r-1}$ ; for  $r = 1$  no members are needed. They meet  $Z$  properly at each successive intersection, because no positive-dimensional component is contained in the base locus. Thus  $Z \cdot E_1 \cdots E_{r-1}$  is an effective zero-cycle. At each  $x \in S$ , the local equations of  $Z, E_1, \dots, E_{r-1}$  form a system of parameters in the regular local ring  $\mathcal{O}_{V,x}$ . Their local intersection multiplicity is the positive length of the quotient by these equations. Each of the  $q^r$  points of  $S$  contributes at least one to the intersection cycle, so

$$q^r \leq \deg(Z \cdot E_1 \cdots E_{r-1}) = mq^{r-1}\delta;$$

see [16, Sections 2.3–2.4]. Dividing by  $q^{r-1}$  proves the claim.  $\square$

*Proof of Theorem 1.1.* Write  $q > 1$  for the algebraic degree of  $f$ , and put  $r = \dim V$  and  $\delta = \deg V$ . If  $r = 0$ , then  $V$  is a point; if  $r = n$ , then  $V = \mathbf{P}^n$ . Assume  $1 \leq r < n$  and suppose, for a contradiction, that  $\delta > 1$ .

Fix an injection  $\omega_V(r+1) \hookrightarrow \mathcal{O}_V(m)$  as in Lemma 5.2. Replace  $f$  by an iterate, still denoted by  $f$ , whose algebraic degree  $q$  satisfies

$$q > \max\{r+1, m\delta\}.$$

Put  $g = f|_V$  and  $t = rq - r - 1$ . By Lemma 2.1, there is a dense open set  $U \subseteq V_{\text{reg}}$  such that  $g^{-1}(U) \subseteq V_{\text{reg}}$  and  $g^{-1}(U) \rightarrow U$  is étale.

A general codimension- $r$  linear subspace meets  $V$  transversely in  $\delta$  distinct points  $y_1, \dots, y_\delta$  of  $U$ . Indeed, it avoids  $V \setminus U$  by dimension, and Bertini's theorem gives transversality on the smooth locus. Set

$$S_j = g^{-1}(y_j), \quad T = S_1 \sqcup \cdots \sqcup S_\delta.$$

Each  $S_j$  consists of  $q^r$  distinct smooth points. Pulling back the  $r$  linear equations of the chosen subspace gives sections of  $\mathcal{O}_V(q)$  with common zero scheme  $T$ . This zero scheme is reduced because  $g$  is étale over the transverse intersection. By Corollary 2.3, the cohomology hypothesis of Proposition 5.1 holds.

Proposition 4.1 gives a functional on  $H^0(S_1, \mathcal{O}_{S_1}(t))$  with nonzero component at every point of  $S_1$ , annihilating all restricted global sections of  $\mathcal{O}_V(t)$ . Extend it by zero on  $T \setminus S_1$ . Proposition 5.1 then gives a section  $a \in H^0(V, \omega_V(r+1))$  satisfying

$$a(x) \neq 0 \quad (x \in S_1), \quad a(x) = 0 \quad (x \in T \setminus S_1).$$

In particular,  $a$  is nonzero and vanishes on  $S_2$ , which exists because  $\delta > 1$ . Its image in  $H^0(V, \mathcal{O}_V(m))$  is still nonzero and still vanishes on  $S_2$ . Lemma 6.1 gives  $q \leq m\delta$ , contrary to the choice of  $q$ .

Therefore  $\deg V = 1$ . If  $\mathbf{P}^s$  is the linear span of  $V$ , the degree bound for a nondegenerate integral variety gives

$$1 = \deg V \geq s - r + 1.$$

Thus  $s = r$ , and the closed integral subvariety  $V \subseteq \mathbf{P}^r$  of dimension  $r$  is all of  $\mathbf{P}^r$ .  $\square$

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